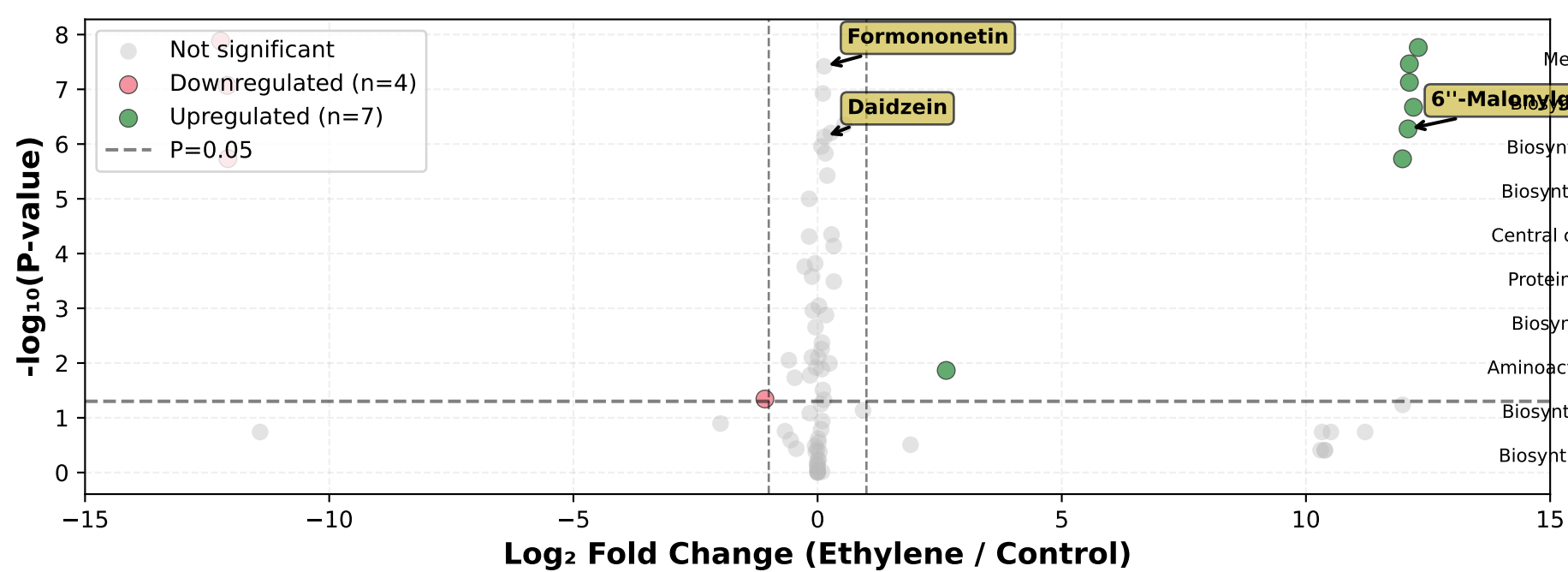
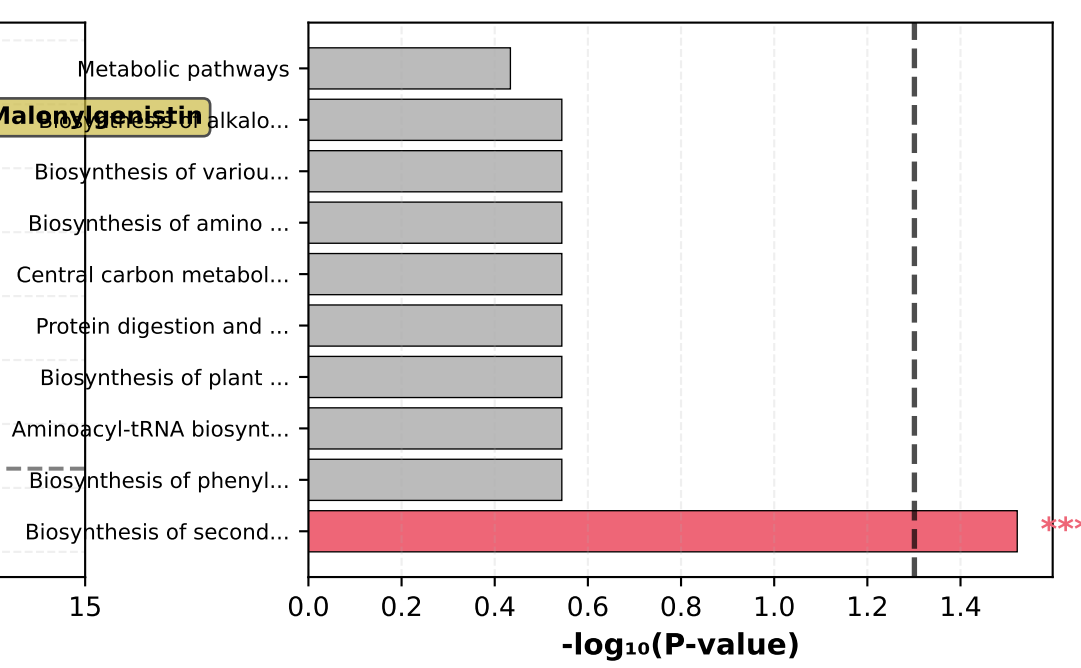


# Ethylene-Induced Metabolic Reprogramming in Soybean Leaves: Coordinated Activation of Isoflavonoid Biosynthesis

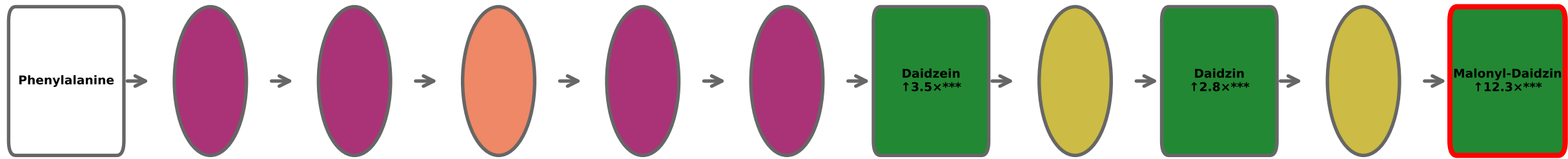
A. Metabolomics: Differential Abundance



B. KEGG Enrichment



C. Isoflavonoid Biosynthesis: Coordinated Metabolite + Enzyme Upregulation



D. Metabolomics Summary

METABOLOMICS

- 79 metabolites quantified
- 43 significant (54%, P<0.05)
- Top hits:
  - Malonyl-daidzin: 12.3x
  - Malonylgenistin: 12.1x
  - Daidzein: P=7.4e-7
  - Formononetin: P=3.8e-8

E. Proteomics Summary

PROTEOMICS

- 6 key enzymes upregulated
- All P<0.05
- Fold changes:
  - IFR: 6.4x
  - CHI: 5.1x
  - 4CL: 3.9x
  - PAL: 3.7x
  - IFS: 3.2x
  - CHS: 2.9x

F. Pathway Analysis

PATHWAY ENRICHMENT

- KEGG map01110:  
P = 0.030 \*\*\* (SIGNIFICANT)  
Biosynthesis of secondary metabolites
- PlantCyc concordance:  
ISOFLAVONOID-SYN  
SECONDARY-METABOLITE-BIOSYNTHESIS